Tide retrospective: full-jet packaging (stage C6)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The quantifier stage the smooth-germ scoping consult listed last and the C5 tide deliberately deferred: upgrade the finite-order recovery to the note's Theorem 3.1 statement shape — *every* partial derivative of positive order.

2 The thing formalised

Two corollaries in `Laplace/OneD/FullJetRecovery.lean`[1]: the finite theorem's coefficient conclusion converted to iterated derivatives (λ = twice the base; higher coefficients cleared of their factorials), and the all-orders form — for smooth admissible losses whose normalized moment data vanishes at every coefficient-sensitive rate, $\partial^k L_1(0) = \partial^k L_2(0)$ for every $k \geq 2$, each order obtained by instantiating the finite theorem at $R = k - 2$. With this, the one-dimensional Theorem 3.1 is formalised in the note's own vocabulary: moment asymptotics determine the full Taylor jet at a nondegenerate minimum, modulo the constant and the vanishing gradient.

3 One catalogue entry

The only friction was the C^∞ grade bookkeeping in current Mathlib: the smoothness exponent lives in WithTop $\mathbb{N}_\infty$, where the coercion of $(\top : \mathbb{N}_\infty)$ is the C^∞ grade and the actual $\top$ is the analytic grade ω; `exact_mod_cast` cannot bridge $\uparrow n \leq \uparrow \top$, and the ∞ notation is scoped. The working idiom is `contDiff_infty.mp`, which converts the C^∞ hypothesis to $\forall n, \mathrm{ContDiff}\ \mathbb{R}\ n\ f$ once and for all.

4 Deliberation and check

Prior-deliberation path (C6 is the scoping consult's last listed stage); definitional conversions only, so no numerical check is feasible — noted per protocol.

5 What remains

The germbij note's one-dimensional story is now complete in Lean at every order. The open commissions are the multivariate Theorem 3.1 (scoping consult in flight as this tide closes) and, further out, the degenerate §7.4 classes beyond dimension one.

[1] `smooth_jet_recovery_iteratedDeriv` and `smooth_full_jet_recovery`.